

1. Operating Systems

Basics

Topic	What to study
OS role	What OS does, kernel vs user space
System calls	fork, exec, wait, open, read, write, close
Process vs program	Program, process, thread
Process states	New, ready, running, waiting, terminated
Context switching	Why it happens, overhead
Kernel mode vs user mode	Why two modes exist

Process Management

Topic	What to study
Process Control Block	PID, state, registers, scheduling info
CPU scheduling	FCFS, SJF, SRTF, Round Robin, Priority
Scheduling metrics	Waiting time, turnaround time, response time
Preemptive vs non-preemptive	Differences and examples
Starvation	What causes it
Aging	How starvation is prevented

Threads and Concurrency

Topic	What to study
Process vs thread	Memory sharing, overhead
User-level vs kernel-level threads	Pros/cons
Multithreading	Benefits and problems
Race condition	What it is, examples
Critical section	Entry, critical, exit, remainder section
Mutex	Lock/unlock

Topic	What to study
Semaphore	Binary and counting semaphore
Spinlock	Where it is used
Condition variable	wait/signal idea
Producer-consumer	Classic synchronization problem
Reader-writer	Classic synchronization problem
Dining philosophers	Deadlock/synchronization problem

Deadlocks

Topic	What to study
Deadlock definition	When processes wait forever
Necessary conditions	Mutual exclusion, hold and wait, no preemption, circular wait
Deadlock prevention	Break one condition
Deadlock avoidance	Banker's algorithm
Deadlock detection	Resource allocation graph
Deadlock recovery	Kill process, preempt resource

Memory Management

Topic	What to study
Logical vs physical address	CPU-generated vs actual RAM address
MMU	Address translation
Contiguous allocation	Fixed/variable partitions
Fragmentation	Internal vs external
Paging	Page, frame, page table
Segmentation	Segment table, base, limit
Paging vs segmentation	Differences
TLB	Translation Lookaside Buffer

Topic	What to study
Virtual memory	Why it is used
Demand paging	Load page only when needed
Page fault	What happens during page fault
Page replacement	FIFO, LRU, Optimal, Clock
Thrashing	Too many page faults
Working set	Active pages of a process

File Systems

Topic	What to study
File attributes	Name, size, permissions, timestamps
File operations	create, open, read, write, close
Directory structure	Single-level, tree, DAG
File allocation	Contiguous, linked, indexed
Inodes	Metadata storage
Free space management	Bitmaps, linked list
Journaling	Crash consistency
Permissions	Read, write, execute

Important OS Interview Questions

You should be able to answer:

- What happens when you run a program?
- What happens during a context switch?
- Process vs thread?
- Mutex vs semaphore?
- Deadlock conditions?
- Paging vs segmentation?
- What is virtual memory?

- What happens on a page fault?
 - What is thrashing?
 - How does fork() work?
 - What happens when you type a command in terminal?
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2. DBMS

Basics

Topic	What to study
DBMS vs file system	Advantages of DBMS
Schema vs instance	Structure vs current data
Data models	Relational, document, key-value
Keys	Super key, candidate key, primary key, foreign key
Constraints	NOT NULL, UNIQUE, CHECK, FOREIGN KEY
Entity-Relationship model	Entity, attribute, relationship
ER diagrams	Cardinality, participation

Relational Model

Topic	What to study
Relation	Table
Tuple	Row
Attribute	Column
Domain	Allowed values
Relational algebra	Selection, projection, join, union, difference

SQL

Topic	What to study
SELECT queries	Filtering, sorting, grouping

Topic	What to study
WHERE	Conditions
GROUP BY	Aggregation
HAVING	Filter after grouping
ORDER BY	Sorting
LIMIT/OFFSET	Pagination
Joins	INNER, LEFT, RIGHT, FULL, CROSS
Subqueries	Nested SELECT
Correlated subquery	Subquery depending on outer query
EXISTS	Existence checking
UNION vs UNION ALL	Duplicate handling
Window functions	ROW_NUMBER, RANK, DENSE_RANK
CTE	WITH clause
Views	Virtual tables
Stored procedures	Basic idea
Triggers	Basic idea

Normalization

Topic	What to study
Functional dependency $X \rightarrow Y$	
Closure	Attribute closure
1NF	Atomic values
2NF	No partial dependency
3NF	No transitive dependency
BCNF	Stronger than 3NF

Topic	What to study
Denormalization	Why sometimes used
Anomalies	Insert, update, delete anomaly

Indexing

Topic	What to study
Index	Why it speeds queries
Primary index	On primary key
Secondary index	On non-primary attribute
Dense vs sparse index	Difference
Clustered index	Data stored physically sorted
Non-clustered index	Separate index structure
B-tree	Balanced search tree
B+ tree	Common DB index structure
Hash index	Equality search
Composite index	Multi-column index
Covering index	Query served from index
Index tradeoff	Faster reads, slower writes

Transactions

Topic	What to study
Transaction	Logical unit of work
ACID	Atomicity, Consistency, Isolation, Durability
Commit	Make changes permanent
Rollback	Undo changes
Schedule	Order of operations
Serial schedule	One transaction after another

Topic	What to study
Serializable schedule	Equivalent to serial
Conflict serializability	Precedence graph
Recoverability	Safe recovery after failure
Cascading rollback	Rollback chain

Concurrency Control

Topic	What to study
Lost update	Two writes conflict
Dirty read	Read uncommitted data
Non-repeatable read	Same read gives different value
Phantom read	New rows appear
Locks	Shared/exclusive
Two-phase locking	Growing and shrinking phase
Strict 2PL	Holds locks till commit
Timestamp ordering	Order by timestamp
MVCC	Multiple versions of data
Isolation levels	Read uncommitted, read committed, repeatable read, serializable

Query Processing

Topic	What to study
Query parsing	Syntax checking
Query optimization	Choose efficient plan
Execution plan	How DB executes query
Join algorithms	Nested loop, hash join, sort-merge join

Topic	What to study
Cost estimation	Based on stats
Materialization	Store intermediate result
Pipelining	Pass result directly

Important DBMS Interview Questions

You should be able to answer:

- Primary key vs unique key?
- Primary key vs foreign key?
- DELETE vs TRUNCATE vs DROP?
- WHERE vs HAVING?
- INNER JOIN vs LEFT JOIN?
- What is normalization?
- 3NF vs BCNF?
- What is indexing?
- Why does index slow down writes?
- B-tree vs B+ tree?
- What are ACID properties?
- What is a transaction?
- What are isolation levels?
- What is deadlock in DBMS?
- What is MVCC?

3. Computer Networks

Basics

Topic	What to study
Network	Connected devices
Protocol	Rules for communication

Topic	What to study
LAN, WAN, MAN	Types of networks
Client-server model	Browser/server example
Peer-to-peer	Direct communication
Bandwidth	Max data transfer rate
Latency	Delay
Throughput	Actual achieved speed

OSI and TCP/IP Models

Layer	What to study
Application	HTTP, DNS, SMTP
Transport	TCP, UDP
Network	IP, routing
Data link	Ethernet, MAC
Physical	Signals, cables

OSI vs TCP/IP Difference between models

Application Layer

Topic	What to study
HTTP	Request-response protocol
HTTPS	HTTP over TLS
HTTP methods	GET, POST, PUT, PATCH, DELETE
HTTP status codes	200, 201, 301, 400, 401, 403, 404, 500
Headers	Metadata
Cookies	Client-side state
Sessions	Server-side state
REST	Resource-based APIs

Topic	What to study
WebSocket	Full-duplex communication
DNS	Domain to IP mapping
SMTP	Email sending
FTP	File transfer

Transport Layer

Topic	What to study
TCP	Reliable, connection-oriented
UDP	Fast, connectionless
TCP vs UDP	Differences and use cases
TCP 3-way handshake	SYN, SYN-ACK, ACK
TCP connection close	FIN, ACK
Flow control	Receiver capacity handling
Congestion control	Network congestion handling
Sliding window	Efficient reliable transfer
Retransmission	Lost packet recovery
Port numbers	Process-level addressing
Socket	IP + port

Network Layer

Topic	What to study
IP address	IPv4/IPv6
Subnetting	Network division
CIDR	Prefix notation
Routing	Path selection
Router	Connects networks

Topic	What to study
NAT	Private to public IP mapping
ICMP	Ping, error messages
ARP	IP to MAC mapping
DHCP	Dynamic IP assignment

Data Link Layer

Topic	What to study
MAC address	Hardware address
Ethernet	LAN protocol
Switch	Connects devices in LAN
Frames	Data link packets
Error detection	CRC
Collision	Shared medium conflict

Security

Topic	What to study
TLS/SSL	Secure communication
Symmetric encryption	Same key
Asymmetric encryption	Public/private key
Digital certificates	Identity verification
Certificate authority	Trusted issuer
Hashing	Integrity
Man-in-the-middle attack	Interception
Firewall	Network filtering
VPN	Secure tunnel

Important CN Interview Questions

You should be able to answer:

- What happens when you enter a URL in browser?
- HTTP vs HTTPS?
- HTTP vs WebSocket?
- TCP vs UDP?
- Explain TCP 3-way handshake.
- Why is TCP reliable?
- What is DNS?
- What is NAT?
- What is subnetting?
- What is a socket?
- What is TLS?
- What happens in HTTPS handshake?
- GET vs POST?
- Cookie vs session?
- 401 vs 403?
- Load balancer vs reverse proxy?

4. OOP and Software Engineering

OOP Basics

Topic	What to study
Class	Blueprint
Object	Instance
Encapsulation	Hide data
Abstraction	Hide implementation
Inheritance	Reuse/extend class
Polymorphism	Same interface, different behavior

Topic	What to study
Constructor	Object initialization
Destructor	Cleanup
Access modifiers	public, private, protected
Static members	Class-level members
Interface	Contract
Abstract class	Partial implementation

Advanced OOP

Topic	What to study
Method overloading	Same name, different parameters
Method overriding	Child class changes parent method
Runtime polymorphism	Virtual functions
Compile-time polymorphism	Function/operator overloading
Multiple inheritance	Diamond problem
Composition	Has-a relationship
Aggregation	Weak has-a
Association	Uses relationship
Coupling	Dependency between modules
Cohesion	Relatedness inside module

SOLID Principles

Principle	Meaning
S	Single Responsibility
O	Open/Closed
L	Liskov Substitution
I	Interface Segregation

Principle Meaning

D Dependency Inversion

Design Patterns

Study basics of:

Pattern	Use
Singleton	One instance
Factory	Create objects
Abstract Factory	Family of objects
Builder	Step-by-step object creation
Strategy	Choose algorithm at runtime
Observer	Event subscription
Decorator	Add behavior dynamically
Adapter	Convert interface
Proxy	Controlled access
Command	Encapsulate request
MVC	Separate model, view, controller

Low-Level Design Problems

Practice these:

1. Parking lot
2. Elevator system
3. Splitwise
4. Library management system
5. Movie ticket booking
6. Chess
7. Snake and ladder
8. Rate limiter

9. Logger system
10. File system
11. ATM
12. Vending machine
13. Cab booking
14. Food delivery
15. Notification system

Important OOP Interview Questions

You should be able to answer:

- OOP vs procedural programming?
- Abstraction vs encapsulation?
- Overloading vs overriding?
- Interface vs abstract class?
- Inheritance vs composition?
- What is polymorphism?
- What are virtual functions?
- What is the diamond problem?
- Explain SOLID principles.
- Explain Factory pattern.
- How would you design a parking lot?
- How would you design Splitwise?

5. Programming Language Fundamentals — C++ Focus

Since you mostly code in C++, prepare this well.

C++ Basics

Topic	What to study
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Compilation process	Preprocess, compile, assemble, link
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Topic	What to study
Header files	Declarations
Namespaces	Avoid name conflicts
References	Alias
Pointers	Address storage
Pointer vs reference	Differences
const	Immutable values
static	Storage duration/class-level
inline	Function expansion hint
typedef/using	Type alias
auto	Type inference

Memory

Topic	What to study
Stack vs heap	Allocation difference
new/delete	Dynamic memory
malloc/free	C-style allocation
Memory leak	Unfreed memory
Dangling pointer	Pointer to freed memory
Smart pointers	unique_ptr, shared_ptr, weak_ptr
RAII	Resource acquisition is initialization
Shallow copy	Copy pointer only
Deep copy	Copy actual resource
Rule of 3/5/0	Copy/move/destructor design

OOP in C++

Topic	What to study
Constructor/destructor Lifecycle	
Copy constructor	Copy object
Assignment operator	Assign object
Move constructor	Move resource
Virtual function	Runtime dispatch
Pure virtual function	Abstract class
Virtual destructor	Important for base class
vtable/vptr	Runtime polymorphism mechanism
Friend function/class	Access private members
Operator overloading	Custom operators

STL

Topic	What to study
vector	Dynamic array
string	Character sequence
pair/tuple	Group values
stack/queue/deque	Linear containers
priority_queue	Heap
set/multiset	Ordered tree
map/multimap	Ordered key-value
unordered_set/map	Hash table
iterator	Container traversal
lower_bound/upper_bound	Binary search
sort/stable_sort	Sorting

Topic	What to study
custom comparator	Sorting custom logic
lambda	Anonymous function
bitset	Fixed-size bits

STL Internals

Container	Internal idea
vector	Dynamic array
deque	Blocked array
stack	Adapter
queue	Adapter
priority_queue	Binary heap
set/map	Red-black tree
unordered_map	Hash table
string	Dynamic char array

Important C++ Interview Questions

You should be able to answer:

- Pointer vs reference?
- Stack vs heap?
- malloc vs new?
- delete vs free?
- What is RAI?
- What is a memory leak?
- What are smart pointers?
- unique_ptr vs shared_ptr?
- vector vs array?
- map vs unordered_map?

- set vs unordered_set?
- How does unordered_map work?
- What happens during vector resizing?
- What is virtual function?
- Why virtual destructor?
- What is copy constructor?
- What is move semantics?
- What is the rule of 3/5/0?

6. System Design Basics

For fresher/M.Tech placements, learn basic system design.

Core Concepts

Topic	What to study
Client-server architecture	Basic web system
API design	REST endpoints
Load balancer	Distribute traffic
Reverse proxy	Gateway before servers
Caching	Redis/CDN/browser cache
Database replication	Primary-replica
Database sharding	Split data
Horizontal scaling	Add more machines
Vertical scaling	Bigger machine
Message queue	Async processing
Rate limiting	Prevent abuse
Consistency	Strong/eventual
Availability	System uptime

Topic	What to study
CAP theorem	Consistency, availability, partition tolerance
Fault tolerance	Survive failures
Idempotency	Safe retries
Backpressure	Handle overload
Observability	Logs, metrics, traces

Storage

Topic	What to study
SQL vs NoSQL	When to use what
Key-value store	Redis/Dynamo style
Document DB	MongoDB style
Column DB	Analytics workloads
Object storage	S3-like storage
Blob storage	Images/videos/files
CDN	Static content delivery

Common Designs

Practice:

1. URL shortener
2. Chat app
3. File upload system
4. Notification system
5. News feed
6. Search autocomplete
7. Rate limiter
8. E-commerce order system
9. Video streaming basic system

10. Job queue/task scheduler

Important System Design Questions

You should be able to answer:

- How do you scale a backend?
 - SQL vs NoSQL?
 - What is caching?
 - What is cache invalidation?
 - What is a load balancer?
 - What is a message queue?
 - What is eventual consistency?
 - What is CAP theorem?
 - How would you design a URL shortener?
 - How would you design a chat app?
 - How would you design a notification system?
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7. Software Engineering and Git

Git

Topic	What to study
git init/clone/status	Basics
add/commit/push/pull	Daily usage
branch	Parallel work
merge	Combine branches
rebase	Replay commits
conflict resolution	Fix merge conflicts
stash	Temporarily save work
reset	Move HEAD

Topic	What to study
revert	Undo safely
cherry-pick	Pick specific commit
pull request	Code review workflow

Testing

Topic	What to study
Unit testing	Test small function
Integration testing	Test combined modules
End-to-end testing	Test full flow
Mocking	Fake dependency
Test coverage	How much code tested
Regression testing	Ensure old bugs do not return

Clean Code

Topic	What to study
Naming	Clear variable/function names
Modular code	Small functions
Error handling	Fail gracefully
Logging	Debug production issues
Code review	Review for correctness
Refactoring	Improve structure without behavior change
Documentation	README, comments, API docs

DevOps Basics

Topic	What to study
Linux commands	ls, cd, grep, find, chmod, ps, kill
Shell scripting	Basic bash

Topic	What to study
Environment variables	Config management
Docker basics	Containerization
CI/CD basics	Automated build/test/deploy
Deployment	Server/cloud basics

8. Web Development Fundamentals

Even for SDE roles, know backend/web basics.

Backend

Topic	What to study
REST API	Resource-based API
JSON	Data format
Authentication	Login identity
Authorization	Permission checking
JWT	Token auth
OAuth	Third-party login
Middleware	Request processing layer
CRUD	Create/read/update/delete
Pagination	Limit large results
Validation	Check input
Error codes	API error handling
Logging	Request/error logs

Frontend Basics

Topic	What to study
HTML/CSS/JS basics	Web foundation

Topic	What to study
DOM	Page structure
React basics	Components, props, state
State management	Local/global state
API calls	fetch/axios
Responsive UI	Mobile-friendly design

Security Basics

Topic	What to study
SQL injection	Malicious SQL input
XSS	Injected script
CSRF	Forged request
Password hashing	bcrypt/argon2
Rate limiting	Prevent abuse
Input validation	Never trust user input
HTTPS	Encrypted traffic